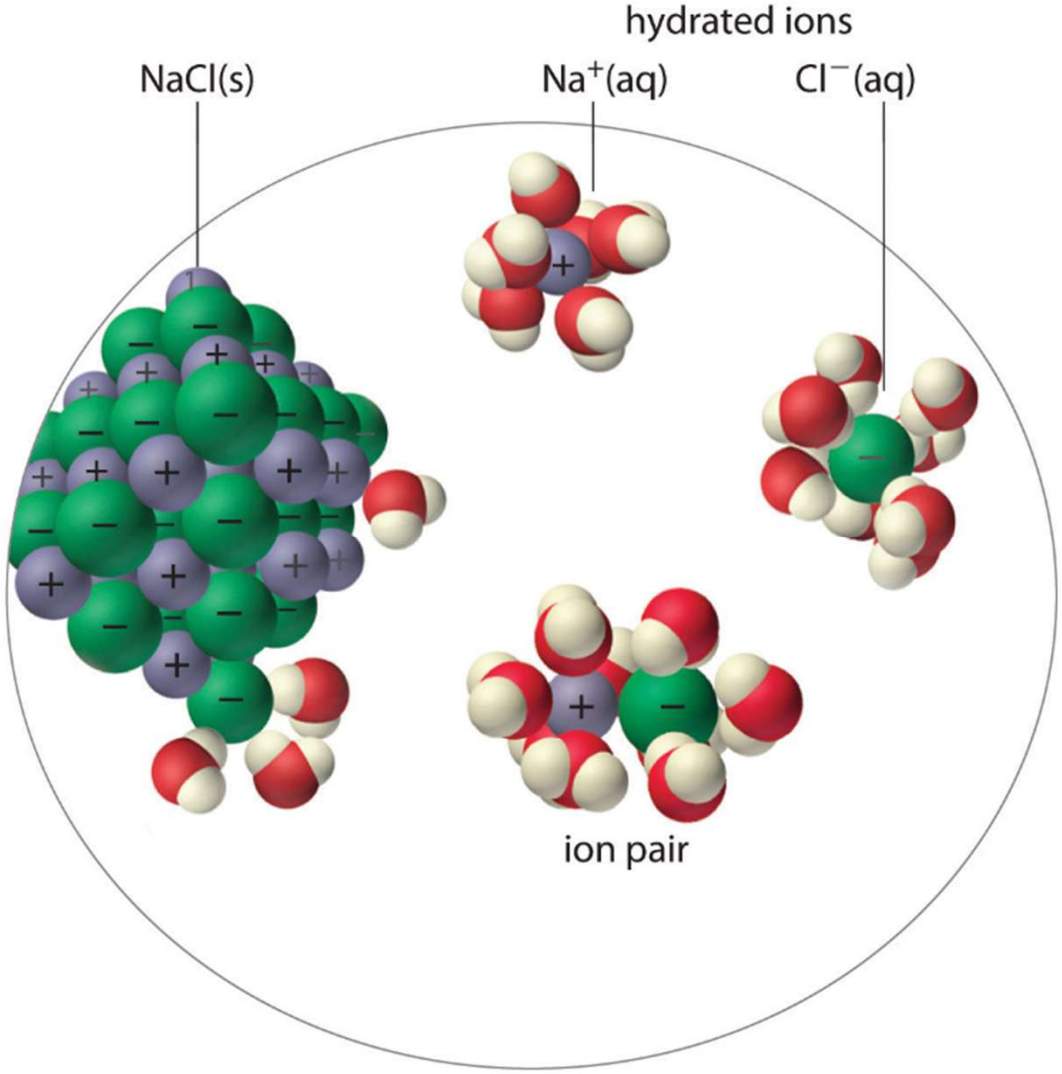
Solubility and Complexing

The solubility of minerals in aqueous solutions are essential for our understanding of fluid properties, mass transport and growth/dissolution processes of minerals in Earth's crust and upper mantle.

Complexing is a key mechanism for mobilising ions (particularly metal ions) in solution.

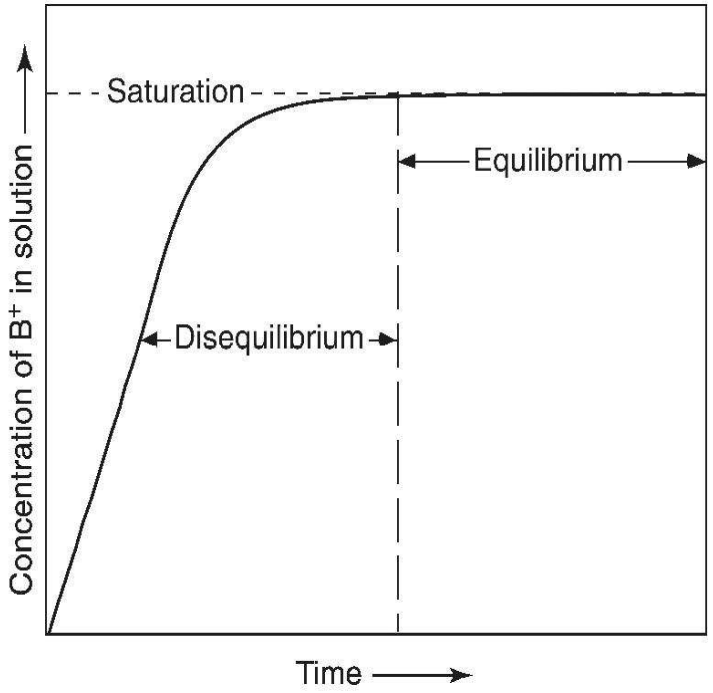


Solubility of Salts

Salt – a compound that contains a cation other than H^+ and an anion other than OH^- or O^{2-}

When a solid salt is placed in water some of it will dissolve – two step process: break-up of ionic lattice then attachment of water molecules to the ions

$$B_pA_{n\,(s)} \leftrightharpoons pB^+_{(aq)} + nA^-_{(aq)}$$



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Concentration dissolved:

- Solubility – as moles
  - moles per Kg water – molality (m)
  - moles per litre water – molarity (M)
- Solubility – as weight
  - grams per Kg water
  - parts per million, ppm (grams/Kg × 10^6)
  - parts per billion, ppb (grams/Kg × 10^9)
- To convert moles to grams × by gram formula weight
- To convert grams to moles ÷ by gram formula weight

Solubility product

Applying the Law of Mass Action to express solubility in water the equilibrium constant K is equivalent to the solubility product K_{sp}

$$B_{\mathrm{p}}A_{\mathrm{n}} \Leftrightarrow \mathrm{p}B^{+} + \mathrm{n}A^{-}; \qquad K_{sp} = \left[B^{+}\right]^{\mathrm{p}}\left[A^{-}\right]^{\mathrm{n}}$$

- Note that the concentration of the solid salt is not included in the K_{sp} expression, its concentration does not change the equilibrium.
- For soluble ionic salts we use ***activity*** in place of ***concentration***, the latter applies for slightly soluble salts.
- K_{sp} are a function of **temperature, pressure** and **pH**. K_{sp}> 0.1 soluble; <0.1 slightly soluble.

e.g. $Ca(SO_4).2H_2O \leftrightharpoons Ca^{2+} + SO_4^{2-} + 2H_2O$

$K_{sp} = [Ca^{2+}][SO_4^{2-}] = 2.51 \times 10^{-5}$

NB It is convention <u>not</u> to report the units for K_{sp} (although some people do, they are effectively moles/L)

Solubility of gypsum ($CaSO_4.2H_2O$)

- Write the dissolution reaction:

$$Ca(SO_4).2H_2O \leftrightharpoons Ca^{2+} + SO_4^{2-} + 2H_2O$$

- K_{sp} at standard temperature and pressure is 2.51×10^{-5}
- At equilibrium the law of mass action applies:

$$K_{sp} = [Ca^{2+}][SO_4^{2-}] = 2.51 \times 10^{-5}$$

- Assume solution is ideal (dilute) with $\gamma Ca^{2+} = \gamma SO_4^{2-} = 1$; x moles of gypsum dissolve to form x moles of Ca^{2+} and x moles SO_4^{2-}

$$K_{sp} = x \times x = x^2$$

Solubility of gypsum = $\sqrt{2.51\times10^{-5}} = 5.01\times10^{-3}$ mol/L

$= 172.2 \times (5.01 \times 10^{-3}) = 0.9$ g/L

Calculating solubilities: another example

- $Al_2(SO_4)_3 \leftrightharpoons 2Al^{3+} + 3SO_4^{2-}$
- $K_{sp} = [Al^{3+}]^2 [SO_4^{2-}]^3$
- $K_{sp} = 69.9$ (at standard temperature and pressure, i.e. 0°C and 1 bar)
- The concentration of the ions are equal to their activities in solution:

$$Al_2(SO_4)_3 \leftrightharpoons 2Al^{3+} + 3SO_4^{2-}$$

- Assume solution is ideal (i.e. dilute) with $\gamma Al^{3+} = \gamma SO_4^{2-} = 1$
- Dissolution of x moles of $Al_2(SO_4)_3$ produces $3x$ moles of SO_4^{2-} and $2x$ moles of Al^{3+}

$$K_{sp} = (2x)^2(3x)^3$$

$$K_{sp} = 2x \times 2x \times 3x \times 3x \times 3x = 108x^5 = 69.9$$

$$= (69.9/108)^{\wedge 0.2} \text{ therefore } x = 0.9147$$

- So a saturated solution of $Al_2(SO_4)_3$ contains:

$$2x = 1.829 \text{ mol/L } Al^{3+} \text{ and } 3x = 2.744 \text{ mol/L } SO_4^{2-}$$

- The solubility of $Al_2(SO_4)_3$ is 0.9147 mol/L (all at STP)

$$= 0.9147 \times 342 = 313.6 \text{ g/L}$$

Ion Activity Product (IAP)

- In natural waters, cations and anions are not normally found in stoichiometric balance (i.e., the ratios in chemical formulas).

- To determine whether an aqueous solution is undersaturated, saturated, or supersaturated with respect to a particular mineral, calculate the IAP.

- To do so, simply substitute the concentrations of the dissolved ions into the Law of Mass Action for that dissolution (remember that the solid has a concentration/activity of 1).

- Then compare the IAP to the K_{sp}.
  - the IAP is the actual condition
  - K_{sp} is the condition for saturation

Ion Activity Product

For example Ganges River (data in Table from last week's lecture):

$[Ca^{2+}] = 0.634 \times 10^{-3}$ m
$[SO_4^{2-}] = 0.088 \times 10^{-3}$ m

$IAP = [0.643 \times 10^{-3}]\,[0.088 \times 10^{-3}] = 56.58 \times 10^{-9}$
$<< K_{sp} = 2.51 \times 10^{-5}$

$IAP > K_{sp}$	supersaturated, precipitation occurs
$IAP \sim K_{sp}$	saturated, no precipitation or dissolution
$IAP < K_{sp}$	undersaturated, dissolution occurs

Sometimes the relationship between IAP and K_{sp} is expressed as a ratio and is known as the ***mineral saturation index***.

Note we haven't discussed kinetics – is it energetically favourable?

How much gypsum will precipitate?

If groundwater has 5×10^{-2} mol/L Ca^{2+} and 7×10^{-3} mol/L SO_4^{2-} then should gypsum precipitate at equilibrium, and how much?

IAP = $(5\times10^{-2}) \times (7\times10^{-3}) = 3.5\times10^{-4}$ ($K_{sp} = 2.51\times10^{-5}$) **YES**

If x mol/L of gypsum precipitate, then at equilibrium:

$[Ca^{2+}] = 5\times10^{-2} - x$; $[SO_4^{2-}] = 7\times10^{-3} - x$

and $[Ca^{2+}][SO_4^{2-}] = 2.51\times10^{-5}$

$$(5\times10^{-2} - x)(7\times10^{-3} - x) = 2.51\times10^{-5}$$

$$x^2 - (5.7\times10^{-2})x + (3.25\times10^{-4}) = 0$$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

After solving for the quadratic equation:

$x_1 = 5.06\times10^{-2}$ mol/L; $\boldsymbol{x_2 = 6.43\times10^{-3}}$ **mol/L**

We choose x_2 because the x_1 causes $[SO_4^{2-}]$ to be negative.

So 6.43×10^{-3} mol/L of gypsum precipitates, and remaining ions dissolved:

and $[Ca^{2+}] = (5\times10^{-2}) - (6.43\times10^{-3}) = 4.36\times10^{-2}$ mol/L

$[SO_4^{2-}] = (7\times10^{-3}) - (6.43\times10^{-3}) = 5.7\times10^{-4}$ mol/L

Evaporation effect

- $[Ca^{2+}]/[SO_4^{2-}]$ increases with precipitation of gypsum

  Before precipitation: $(5\times10^{-2})/(7\times10^{-3}) = 7.1$

  After precipitation: $(4.36\times10^{-2})/(5.7\times10^{-4}) = 76$

- 6.43×10^{-3} mol/L x 172.2 = 1.11g of gypsum precipitates

- Precipitation not only reduces the concentrations of ions it changes the chemical composition.

- This process occurs when salts precipitate during evaporative concentration such as in a desert lake

Evaporation effect

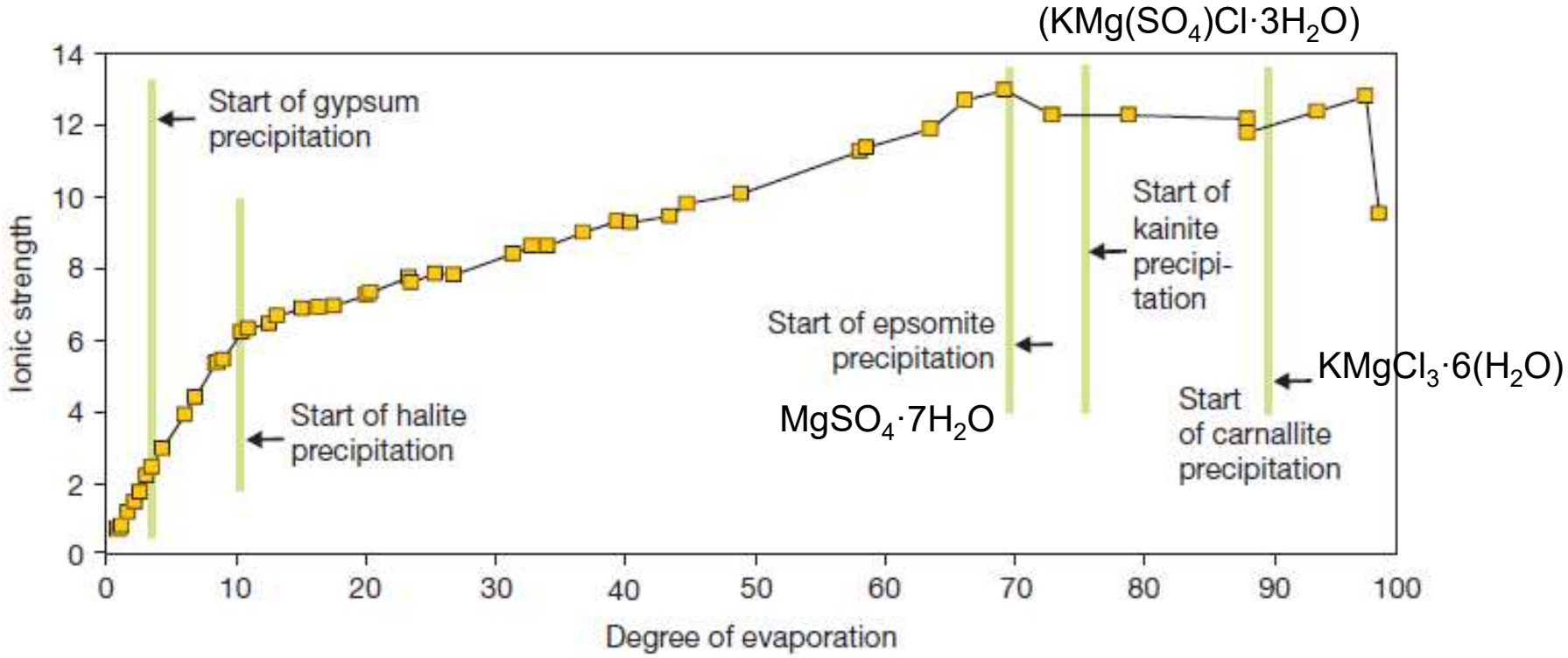


Figure 2 Ionic strength of the evaporating Caribbean seawater brines, based on data by McCaffrey et al. (1987).

- The evaporation path of high ionic strength solutions like seawater cannot be predicted using solubility data alone. There are ion pairs formed by interaction of cations and anions in solution.

The common ion effect

- Natural waters are complex because they contain mixed anion and cation species.
- They may potentially precipitate several different salts.
- Consider an aqueous solution saturated with respect to gypsum and barite ($BaSO_4$).

Here we can write two equilibria that share a **common ion**:

$$Ca(SO_4).2H_2O \leftrightharpoons Ca^{2+} + SO_4^{2-} + 2H_2O$$

$$BaSO_4 \leftrightharpoons Ba^{2+} + SO_4^{2-}$$

The common ion effect

$$Ca(SO_4).2H_2O \leftrightharpoons Ca^{2+} + SO_4^{2-} + 2H_2O$$
$$BaSO_4 \leftrightharpoons Ba^{2+} + SO_4^{2-}$$

Applying the Law of Mass Action we obtain:

$$K_{sp}^{CaSO_4} = \left[Ca^{2+}\right]\left[SO_4^{2-}\right] = 2.51\times10^{-5}$$

$$K_{sp}^{BaSO_4} = \left[Ba^{2+}\right]\left[SO_4^{2-}\right] = 1.0\times10^{-10}$$

SO_4^{2-} must have the same activity in both equilibria so we can eliminate it to give:

$$\frac{\left[Ca^{2+}\right]}{\left[Ba^{2+}\right]} = \frac{2.51\times10^{-5}}{1\times10^{-10}} = 251{,}189$$

The concentration (activity) of Ca^{2+} is ~250,000 times higher than that of Ba^{2+} when the solution is saturated with respect to gypsum and barite.

Common ion effect

To determine the SO_4^{2-} concentration

From earlier the K_{sp} are:

$$[Ca^{2+}][SO_4^{2-}] = 2.51 \times 10^{-5}$$
$$[Ba^{2+}][SO_4^{2-}] = 1.0 \times 10^{-10}$$

Because the solubility of $CaSO_4$ is >> $BaSO_4$ and $[Ca^{2+}] >> [Ba^{2+}]$

Then approximate $[Ca^{2+}] + [Ba^{2+}] \sim [Ca^{2+}]$

So: $[Ca^{2+}] \sim [SO_4^{2-}]$

Since $K_{sp} = [Ca^{2+}][SO_4^{2-}] = 2.51 \times 10^{-5}$

We showed earlier (slide 4) that: $[Ca^{2+}] = [SO_4^{2-}] = 5.01 \times 10^{-3}$ mol/L

Subst. this concentration: $[Ba^{2+}][SO_4^{2-}] = 1 \times 10^{-10}$

$[Ba^{2+}] = 1 \times 10^{-10} / 5.01 \times 10^{-3}$

In presence of $CaSO_4$ **$[Ba^{2+}] = 2.0 \times 10^{-8}$** mol/L

In the absence of $CaSO_4$ **$[Ba^{2+}] = 1.0 \times 10^{-5}$** mol/L

When sharing a common ion the solubility of Ba^{2+} suppressed by presence of Ca^{2+}

Replacement minerals

- If a solution saturated in barium sulphate comes in to contact with gypsum, then the calcium sulphate can dissolve and therefore increase the sulphate concentration.

- This means the solution becomes ***supersaturated*** in barium sulphate and barite will begin to precipitate and “replace” the gypsum.

- This continues until the $[Ca^{2+}]/[Ba^{2+}]$ ratio reaches 250,000 required for equilibrium, then the replacement processes ceases.

- Replacement of one mineral with another is extremely common in nature and affects sulphates, carbonates and sulphides. Ore mineralogy.

- In each case introduction of a *common ion* (in our example sulphate anions) causes the solution to become supersaturated in the less soluble compound (in our example barium sulphate).

- The rule is that the *less soluble* compound always replaces the *more soluble* one - pseudomorphs

Ion Complexing

Most fresh (meteoric) waters (stream, river, lake and groundwaters) waters are dilute (ionic strength, I<0.1) the ions are highly dissociated.

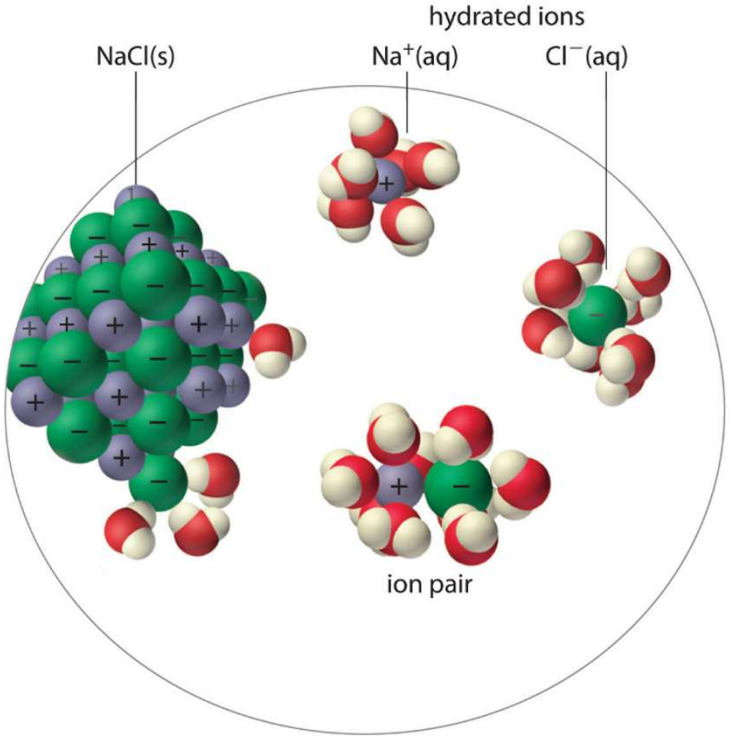
$$I = \frac{1}{2}\sum m_i z_i^2$$

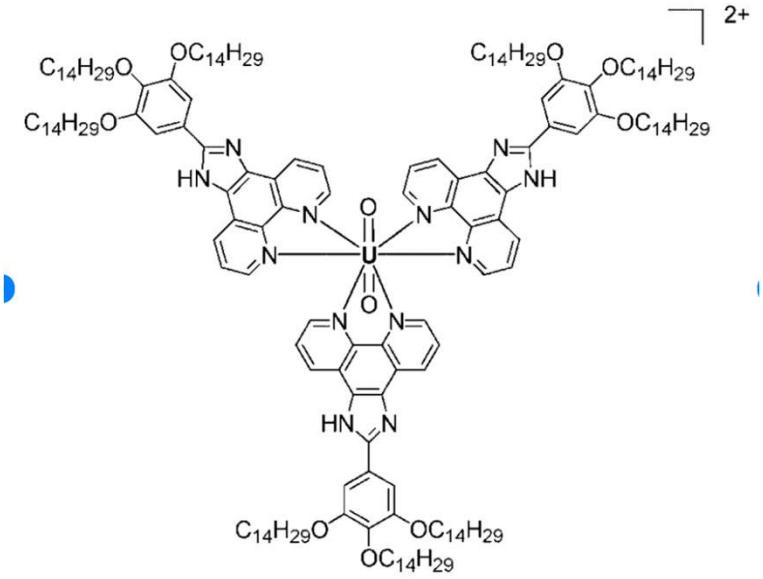
In concentrated natural waters like seawater (I=0.7) there is interaction between cations and anions causing the molecules to form groups or clusters, these are known as ***complex ions.*** This is a very effective way to mobilise (particularly) metal ions.

Complex ion formation defined as: $K_n = \frac{[ML]}{[M][L]}$ where M is metal concentration and L is ligand concentration

Ion Complexing

- **Ion pairs** - neutral molecules composed of two ions of opposite charge (e.g. $CaCO_3$ and $CaSO_4$) – weak bonding low stability.
- **Coordination complexes** – composed of a central atom surrounded by anions or neutral species (ligands). E.g. $Cu(H_2O)_6^{2+}$
- **Chelates** – complexes between metal ions and dissolved anions or organic matter (ligands or chelating agent). Metal chelates are important for mobilising metals in soils and for the uptake of metals in plants.





Complex ions: Gold extraction

- Complex ion formation is used to extract **gold** from low-grade gold containing rock. Gold is very unreactive.
- A small concentration of Au+ ions are formed by oxidation of Au. This oxidation takes place in the presence of CN^- ions (usually as NaCN)
- The formation constant of $Au(CN)_2^-$ is very large.
- The Au is oxidized by bubbling air through a suspension of Au containing ore in the presence of CN^- :

$$4Au(s) + 8CN^-(aq) + O_2(g) + 2H_2O(l) \leftrightharpoons 4Au(CN)_2^-(aq) + 4OH^-(aq)$$

- The pH of the resulting slurry is raised by adding lime (CaO) or another alkali to ensure that cyanide ions do not change into toxic hydrogen cyanide gas.
- Gold recovered onto carbon or by electrolysis

Example: Aluminium production

- Aluminum ore contains Fe_2O_3 impurities along with aluminum oxide, Al_2O_3.
- A strong base is added to dissolve the Al_2O_3 as the complex ion $Al(OH)_4^-$.
- The iron (III) ion does not form a complex ion with hydroxide, therefore the Fe_2O_3 does not dissolve.
- The resulting solution is filtered and acid is added to the filtrate to precipitate $Al(OH)_3$.
- The filtrate must not be made too acidic or the $Al(OH)_3$ will re-dissolve.
- The mixture solid impurities 40% Fe_2O_3 is called red mud







Summary - 1

- The reaction between an acid and a base produces a **salt** and **water**.
- The **solubility** of the salt in water depends upon a number of factors including T, P, pH and activity.
- The solubility can be expressed by applying the Law of Mass Action to obtain a **solubility product K_{sp}**. The higher the K_{sp} the more soluble the salt.
- The **Ion Activity Product** is a simple calculation that allows geochemists to assess whether a given mineral is close to or far from equilibrium with its solubility products.
- Introduction of a **common ion** causes the solution to precipitate the **less soluble** salt, such **mineral replacement** reactions are common in nature

Summary - 2

- In **concentrated** solutions like seawater and brines, the **met**al content of solution can increase by **complexing**, this is important in the extraction of metals from mineral ores and has environmental consequences.

Reading – Chapter 7.4 - Misra